

AZAM HUSSAIN

Junior AI/ML Engineer | Generative AI | RAG | LLMs | NLP

+92 312 6014216 | azamsindhu13@gmail.com | Lahore, Pakistan

LinkedIn: <https://linkedin.com/in/azam-hussain-681695325> | GitHub: <https://github.com/azam-hussain-ml> | Portfolio: <https://azam-portfolio-pi.vercel.app/>

PROFESSIONAL SUMMARY

Junior AI/ML Engineer specializing in LLMs, Retrieval-Augmented Generation (RAG), NLP, and multimodal AI, with hands-on experience building and deploying 3 end-to-end AI systems. Delivered ML models achieving 89% accuracy, 87% F1, 95% precision, and 91% recall, while reducing fraud false negatives by 22%. Skilled in Python, LangChain, ChromaDB, OpenAI/Gemini APIs, REST API integration, LLM evaluation, and AI application deployment. Microsoft Azure AI-900 certified.

TECHNICAL SKILLS

Generative AI & RAG: LLMs, RAG, prompt engineering, prompt/LLM evaluation, embeddings, semantic search/retrieval, vector databases (ChromaDB), document chunking, grounded generation, multimodal AI

ML & Deep Learning: TensorFlow, Keras, PyTorch, scikit-learn, transformer models, CNNs, fine-tuning, transfer learning, feature engineering, data preprocessing, model benchmarking, model evaluation, SMOTE

Backend & APIs: Python, FastAPI, Flask, Streamlit, REST API design/integration, OpenAI API, Google Gemini, Azure OpenAI, Hugging Face, Apify API

Data, Deployment & Cloud: NumPy, Pandas, SQL fundamentals, data pipelines, Docker, Git/GitHub, secrets management, environment/dependency management, reproducible builds, Microsoft Azure, Azure ML, Cognitive Services

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern - FlyRank AI | 2026 - Present

- Shipped LLM-powered product functionality from prototype to working implementation by owning OpenAI API integration and prompt design, enabling end-to-end integration into the broader system.
- Established a held-out test-query evaluation set and pass-rate metric, replacing subjective prompt tuning with repeatable, evidence-based LLM evaluation and measurable prompt iteration.
- Converted an ambiguous business brief into a measurable ranking specification by defining the target variable, feature set, and evaluation metric used to guide a content opportunity scoring model.

AI/ML Intern - CareerCore, Lahore | 2026 | 3 months

- Delivered 3 end-to-end ML models in 3 months: a 7-class CNN skin-disease classifier at 89% accuracy, an NLP sentiment analyzer at 87% F1 across 50,000 reviews, and a fraud detector at 95% precision and 91% recall on 284,807 transactions.
- Improved sentiment F1 by 4-6% by benchmarking Logistic Regression, Naive Bayes, and SVM, while reducing fraud false negatives by 22% with SMOTE on a dataset with 99.83% class imbalance.
- Reduced image-classifier overfitting by 18% and reached 89% accuracy on 10,000 labelled images by switching from training from scratch to MobileNet transfer learning and fine-tuning.

SELECTED GENERATIVE AI PROJECTS

RAG-Based Knowledge Management System | Live: <https://huggingface.co/spaces/azam8954/rag-knowledge-management-system>

- Built and deployed a 5-stage RAG pipeline over private documents: ingestion, chunking, embedding generation, ChromaDB vector-database retrieval, and grounded source-cited LLM response generation via Flask.
- Improved semantic retrieval by evaluating chunk size and overlap against a fixed question set; resolved deployment and security blockers involving LangChain changes, SSL/environment failures, and committed credentials using git filter-repo.

VisionVoice AI - Multimodal Assistant | Live: <https://visionvoice-ai.streamlit.app/>

- Integrated 4 multimodal AI capabilities in one Streamlit interface: Whisper speech-to-text, GPT-4o-mini conversational chat, text-to-speech, and image generation.
- Resolved transcription and model-migration issues by routing English audio through the Whisper translations endpoint and migrating image generation from DALL-E 3 to gpt-image-1 after upstream model retirement.

Generative AI Job Recommender | Live: <https://azam-ai-job-recommender.streamlit.app/>

- Built a 2-stage job-matching workflow that ingests live listings through the Apify API, then uses an LLM to match and rank roles against a parsed CV.
- Profiled inference cost and response latency per query before migrating from OpenAI to Gemini, reducing API spend while maintaining output quality on a manual review set.

EDUCATION & CERTIFICATIONS

MSc Computer Science - The Superior University, Lahore | 2022

Microsoft Azure AI Fundamentals (AI-900) - Microsoft, 2026 | Credential ID: 9ED38B7EB43D0324

AI Fluency & Claude Code 101 - Anthropic Academy | AI / ML / Deep Learning Programme - NAVTTC, Pakistan